🍽️ Restaurant Recommendation System

1. Introduction

The **Restaurant Recommendation System** is a web-based application that suggests restaurants to users based on their preferences such as restaurant name and location.

It uses **Machine Learning (TF-IDF + Cosine Similarity)** to analyze user reviews and recommend similar restaurants.

2. Objectives

- Provide personalized restaurant recommendations
- Use text-based similarity (reviews) for better accuracy
- Allow users to search by **restaurant name + location**
- Automatically detect user location using GPS
- Create a simple and user-friendly interface

3. System Architecture

3.1 Frontend

- HTML (templates)
- CSS (basic styling)
- Flask Jinja templates

3.2 Backend

- Python (Flask Framework)

3.3 Machine Learning

- TF-IDF Vectorizer
- Cosine Similarity

3.4 Dataset

- Zomato Dataset (`zomato.csv`)

5. Technologies Used

Category	Technolgy
Programming langauge	Python
Framework	Flask
ML Libraries	ScikitLearn
Data Handling	Pandas
Visualization	Matplotlib
API	Open Sreet Map

6. Working of the System

Step 1: Data Loading

- Dataset is loaded using Pandas
- Unnecessary columns are removed

Step 2: Data Cleaning

- Remove null values
- Remove duplicates
- Convert ratings to numeric
- Clean text (remove punctuation, lowercase)

Step 3: Feature Extraction

- Use **TF-IDF Vectorizer** on `reviews_list`

Step 4: Similarity Calculation

- Use **Cosine Similarity** to compare restaurants

Step 5: Recommendation Logic

- Filter restaurants by location
- Find matching restaurant name
- Get top similar restaurants
- Remove duplicates
- Return top 10 results

6. Location Feature

- Uses **OpenStreetMap Reverse Geocoding API**
- Converts latitude & longitude → location name
- Helps auto-fill user location

7. API Endpoints

Route	Method	Description
/	GET	Homepage
/reccomendation	GET/POST	Reccomendation Page

8. User Flow

- User opens homepage
- Navigates to recommendation page
- Enters:
    - Restaurant name
    - Location (or auto-detected)

- System processes input
- Displays recommended restaurants

9. Features

- Content-based recommendation
- Location filtering
- Duplicate removal
- Clean UI with Flask
- Auto location detection

10. Limitations

- Dataset limited to available data
- No real-time Zomato API integration
- Recommendations depend on text similarity only
- Not using deep learning (basic ML only)

11. Future Enhancements

- Add **collaborative filtering**
- Use **deep learning (BERT / NLP models)**
- Integrate real-time APIs
- Add user login & history
- Deploy on cloud (AWS / Render / Railway)

12. How to Run the Project

Step 1: Create Virtual Environment

```
python -m venv pyenv
```

Step 2: Activate

```
.\pyenv\Scripts\Activate
```

Step 3: Install Dependencies

```
pip install flask pandas scikit-learn requests
```

Step 4: Run App

```
python app.py
```

Step 5: Open Browser

http://127.0.0.1:5000/

13. Conclusion

The **Restaurant Recommendation System** successfully provides personalized restaurant suggestions using machine learning techniques.

It demonstrates how **NLP + Web Development** can be combined to build real-world applications.